Sofia Montebugnoli

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Bologna, Italy

Short Presentation

I am Sofia Montebugnoli (Graduate Student Member, IEEE; Student Member, ACM), a postdoctoral researcher at the University of Bologna, Italy. I earned my PhD in Computer Science and Engineering there in July 2026 with a dissertation on a cloud-native framework for Open RAN service management and orchestration, which I developed within the PNRR RESTART project in collaboration with Imola Informatica S.p.A. My research covers cloud-native management and orchestration for Open RAN and next-generation mobile networks, including Near-RT RIC platforms, xApp/rApp lifecycle management, and Kubernetes-based orchestration. In 2025 I spent six months as a visiting researcher in Prof. Tommaso Melodia's group at Northeastern University.
I chair the IEEE ComSoc Student Branch Chapter at the University of Bologna and sit on the IEEE ComSoc Young Professionals Webinar Team. From 2024 to 2026 I was Student Member Representative on the IEEE ComSoc EMEA Board. I also work as ICT Research Consultant at Imola Informatica S.p.A. and tutor two university courses.

Research Area and Expertise

Domain Areas: 5G/6G Networks, Open Radio Access Networks (O-RAN), AI-RAN, Smart Cities, Green IT, Industry 4.0/5.0
Technical Expertise: Cloud Continuum, Service Management and Orchestration (SMO), Near-RT RIC, xApp/rApp Development, MLOps, DevOps for Telecommunications, High Performance Computing, Cloud-Native Architectures

Education

University of Bologna ALMA MATER STUDIORUM [🌐] *March 2023 - July 2026*
PhD in Computer Science and Engineering – XXXVIII Cycle Bologna, Italy
- Thesis: "A Cloud Native Solution for Service Management and Orchestration in Next Generation Networks"
- Final grade: Excellent (defended on July 10, 2026)
- Supervisors: Prof. Luca Foschini, Prof. Paolo Bellavista
- Research Group: Mobile Middleware Research Group [🌐]

Northeastern University [🌐] *April 2025 - September 2025*
PhD Visiting Research Period, Institute for the Wireless Internet of Things (WIoT) Boston, MA, USA
- Research Group: WINES Lab, led by Prof. Tommaso Melodia
- Cloud-native deployment and automation of Open RAN systems

University of Bologna ALMA MATER STUDIORUM *September 2020 - December 2022*
Master's Degree (Laurea Magistrale) in Computer Engineering Bologna, Italy
- Final Grade: 110/110 cum laude (with Honors)
- Thesis: "Energy consumption of parallel linear systems solver algorithms on HPC architecture"
- Supervisors: Prof. Anna Ciampolini, Prof. Daniela Loreti

University of Bologna ALMA MATER STUDIORUM *2022*
Advanced Training Course in Industry 4.0 Bologna, Italy

University of Bologna ALMA MATER STUDIORUM *September 2017 - October 2020*
Bachelor's Degree (Laurea Triennale) in Computer Engineering Bologna, Italy
- Thesis: "Progetto e implementazione dell'interfaccia utente per la piattaforma tuProlog"

Francesco Alberghetti Institute *September 2012 - July 2017*
High School Diploma in Applied Sciences Imola, Italy

Academic Third Mission

IEEE Communications Society *September 2026 - Present*
Member, Young Professionals Webinar Team Online
- Identifying speakers and topics for the ComSoc YP series of online technical talks
- Planning joint seminars with other IEEE Societies

IEEE Communications Society *November 2025 - Present*
Student Branch Chapter Chair Bologna, Italy
- Chair for the Student Branch Chapter of IEEE ComSoc at University of Bologna
- Organization of events, seminars, and technical workshops for students in Communication-related topics
- Networking events connecting academic researchers with local industry

IEEE Communications Society *May 2024 - April 2026*
Student Member Representative – EMEA Board Europe, Middle East, Africa
◦ Representative for student members in the Europe, Middle East, and Africa region
◦ Advocated for student interests and promoted IEEE ComSoc activities across EMEA

N²Women (Networking Networking Women) *February 2025 - February 2026*
Social Media Co-Chair and Board Member USA
◦ Managed social media outreach and community engagement for women in networking and communications research

Professional Experience

University of Bologna *March 2026 - Present*
Postdoctoral Researcher Bologna, Italy
◦ Cloud-native management and orchestration for Open RAN, AI-RAN, and next-generation mobile networks
◦ Design of Kubernetes-based orchestration frameworks, service mesh integrations, and intent-driven automation
◦ Scalable resource management for disaggregated 5G/6G RAN infrastructures
◦ Leading system integration, validation, and performance evaluation activities with academic and industrial partners

Northeastern University – WINES Lab *April 2025 - September 2025*
Visiting Researcher Boston, MA, USA
◦ Designed and developed MANATEE, a DevOps platform for the full lifecycle management of Open RAN xApps, covering build, deployment, and automated testing against real and emulated RAN environments
◦ Contributed to AutoRAN, funded by the NTIA Public Wireless Supply Chain Innovation Fund, on zero-touch integration, deployment, and continuous testing of multi-vendor cellular stacks; worked on the cloud-native and infrastructure-as-code components of the automation pipeline

University of Bologna *September 2023 - Present*
Teaching Assistant – Distributed Systems M (Master's Course) Bologna, Italy
◦ Preparation and delivery of practical exercises on DevOps, Jakarta EE, WildFly, Docker, Spring, Spring Boot, and Kubernetes
◦ 30 hours per academic year (2023/24, 2024/25, 2025/26)

University of Bologna *September 2025 - Present*
Teaching Assistant – Computer Networks T (Bachelor's Course) Bologna, Italy
◦ Academic tutor for the Bachelor's course taught by Prof. Antonio Corradi, Prof. Armir Bujari, and Prof. Andrea Sabbioni
◦ 30 hours for academic year 2025/26

Imola Informatica S.p.A. – Technology Transfer *November 2023 - Present*
ICT Research Consultant – ImolaLab R&D Department Imola, Italy
◦ Contribution to R&D IT department supporting technology transfer between academia and industry
◦ Assistance for Bachelor and Master thesis internships bridging university research with industrial applications
◦ Research and evaluation of cloud products and emerging technologies for practical deployment

ECIPAR Regionale for CNA *February 2023 - June 2023*
Engineering Lecturer – Front-End Web Technologies Bologna, Italy
◦ Delivered 60 hours of lectures on HTML, CSS, JavaScript, and React to high-school graduates
◦ Curriculum designed to prepare students for full-stack developer positions

University of Bologna / Injenia S.r.l. *January 2023 - February 2023*
Research Assistant Bologna, Italy
◦ Consulting project on multicloud observability using Terraform, Kubernetes, and ElasticSearch
◦ Developed solutions for text detection and monitoring across cloud platforms

Officine Volta S.p.A. *October 2018 - December 2018*
Marketing Assistant Castel San Pietro Terme, Italy
◦ Updated the 2019 price list with price revisions and graphic redesign using Adobe InDesign, Illustrator, and Excel

Officine Volta S.p.A. *June 2016 - July 2016*
Marketing Assistant Castel San Pietro Terme, Italy
◦ Designed a new price list format and a customized offer proposal using Adobe InDesign, Illustrator, and Photoshop

Emerson Network Power *June 2015 - July 2015*
ICT System Administrator Intern Castel Guelfo di Bologna, Italy
◦ System administration: Ubuntu OS setup, VPN configuration, cloud computing maintenance

PUBLICATIONS

J=JOURNAL, C=CONFERENCE

Author ORCID: 0009-0001-2461-8468

[1] **[C]** S. Montebugnoli, L. Foschini, "A Multicloud Observability Support Based on ElasticSearch for Cloud-native Smart Cities Services," 2023 IEEE Symposium on Computers and Communications (ISCC). IEEE, 2023. DOI: 10.1109/ISCC58397.2023.10218075.

[2] **[C]** S. Montebugnoli, A. Ciampolini, "Energy consumption comparison of parallel linear systems solver algorithms on HPC infrastructure," 2023 The International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC23). ACM, 2023. DOI: 10.1145/3624062.3624266.

[3] **[C]** J. L. Herrera Gonzalez, S. Montebugnoli, D. Scotece, L. Foschini, "xSTART: xApp Simulated Evaluation Environment for Developers," 2024 IEEE International Workshop on Metrology for Industry 4.0 & IoT (MetroInd4.0&IoT2024). IEEE, 2024. DOI: 10.1109/MetroInd4.0IoT61288.2024.10584214.

[4] **[C]** J. L. Herrera, S. Montebugnoli, P. Bellavista and L. Foschini, "Enabling Reusable and Comparable xApps in the Machine Learning-Driven Open RAN," 2024 IEEE 25th International Conference on High-Performance Switching and Routing (HPSR), Pisa, Italy, 2024. DOI: 10.1109/HPSR62440.2024.10635962.

[5] **[C]** S. Montebugnoli, A. Sabbioni, L. Foschini, "Evaluating Mesh Communications in Disaggregated Near-RT RIC for 5G Open RAN: a Functional and Performance Analysis," 2024 IEEE Global Communications Conference (Globecom), Cape Town, South Africa, 2024. DOI: 10.1109/GLOBECOM52923.2024.10901151.

[6] **[J]** J. L. Herrera, S. Montebugnoli, D. Scotece, L. Foschini and P. Bellavista, "A Tutorial on O-RAN Deployment Solutions for 5G: From Simulation to Emulated and Real Testbeds," in IEEE Communications Surveys & Tutorials, 2025. DOI: 10.1109/COMST.2025.3577687.

[7] **[C]** S. Montebugnoli, A. Sabbioni, F. Callegati, L. Foschini, and P. Bellavista, "CLO5ER: a Composable Lightweight Observability for 5G RAN Environment inside Near RT RIC," 2025 IEEE International Conference on Communications (ICC), Montreal, Canada, 2025. DOI: 10.1109/ICC52391.2025.11162022.

[8] **[C]** A. Sabbioni, L. Serfilippi, S. Montebugnoli, A. Bujari, and A. Corradi, "MLCOps: a Platform to Support Cloud Continuum Machine Learning Operations," 30th IEEE Symposium on Computers and Communications (ISCC), Bologna, Italy, 2025. DOI: 10.1109/ISCC65549.2025.11326358.

[9] **[C]** S. Montebugnoli, E. Drudi, A. Sabbioni, G. Di Modica, and L. Foschini, "TORNADO: TOSCA-Enabled Orchestration for RAN Network Functions Automating DevOps in O-Cloud," 30th IEEE Symposium on Computers and Communications (ISCC), Bologna, Italy, 2025. DOI: 10.1109/ISCC65549.2025.11325950.

[10] **[J]** S. Montebugnoli, A. Sabbioni, L. Foschini, and A. Corradi, "Service Mesh: the Rise of Event-Driven Asynchronous Mesh in Cloud Continuum," in IT Professional, vol. 28, no. 1, pp. 51-58, Jan.-Feb. 2026, DOI: 10.1109/MITP.2025.3605101.

[11] **[J]** J. L. Herrera, S. Montebugnoli, P. Bellavista and L. Foschini, "Management and Optimization of xApp and Near-Real-Time RIC Deployment in the Edge," in IEEE Networking Letters. DOI: 10.1109/LNET.2026.3664166.

[12] **[C]** N. Giulianelli, A. Sabbioni, S. Montebugnoli, A. Bujari, and A. Corradi, "KOTO: A Kubernetes-Based Platform for Secure and Scalable OT Orchestration in Industry 5.0," In Proceedings of the 18th IEEE/ACM International Conference on Utility and Cloud Computing (UCC'25). ACM, 2025. DOI: 10.1145/3773274.3774700.

[13] **[C]** N. Giulianelli, A. Sabbioni, S. Montebugnoli, A. Bujari and A. Corradi, "Forging Industry 5.0 DevOps: Towards Secure and Scalable OT Orchestration Powered by Kubernetes," 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC), Las Vegas, NV, USA, 2026. DOI: 10.1109/CCNC65079.2026.11366408.

[14] **[C]** S. Montebugnoli, A. Benenati, A. Sabbioni, and L. Foschini, "Bridging Cloud and Edge for Digital Twins: A Qualitative and Quantitative Study of WASM, Unikernels, and Containers," ICC 2026, IEEE International Conference on Communications, Glasgow, United Kingdom, 2026, pp. 1-6, DOI: 10.1109/ICC59461.2026.11587424.

[15] **[J]** S. Montebugnoli, A. Sabbioni, L. Foschini, P. Bellavista, "PANDA5: A Specialized Kubernetes Control Plane to Support QoS-Aware O-RAN xApp Orchestration," IEEE Communications Magazine. DOI: 10.1109/MCOM.001.2500679.

[16] **[C]** A. Benenati, S. Montebugnoli, I. Guarino, L. Foschini, "DREAMS: Adaptive Orchestration of Network Digital Twin Models at the RAN Edge," 2026 IEEE Global Communications Conference (Globecom), Macau, China, 2026 (Accepted).

Works Under Review

[17] **[J]** S. Zandoli, D. Scotece, S. Montebugnoli, L. Foschini, "TiREx: Tilt Regulation Explainability through xApp with DRL." Submitted to Computer Networks, September 2025.

[18] **[J]** A. Sabbioni, S. Montebugnoli, A. Bujari, P. Bellavista, "I-ORCAS: Intent-based ORchestration for Cloud Continuum Automation via Serverless." Submitted to Springer Nature Link Journal of Cloud Computing, October 2025.

[19] **[J]** S. Montebugnoli, L. Bonati, A. Sabbioni, L. Foschini, P. Bellavista, S. D'Oro, M. Polese, T. Melodia, "MANATEE: A DevOps Platform for xApp Lifecycle Management and Testing in Open RAN." Submitted to IEEE Transactions on Network and Service Management, January 2026. DOI: 10.48550/arXiv.2601.14009.

[20] **[J]** S. Montebugnoli, A. Benenati, A. Sabbioni, L. Foschini, and P. Bellavista, "KOALA: A Digital Twin-driven, SLO-aware orchestrator for 6G AI-and-RAN platforms." Submitted to Computer Networks Elsevier, October 2026. DOI: 10.2139/ssrn.7642217.

[21] **[J]** A. Sabbioni, S. Montebugnoli, C. Chen, L. Foschini, A. Corradi, "DRACO: SLA-Aware Resource Management for Cloud Continuum Control Planes." Submitted to IEEE Software, July 2026.

[22] **[J]** S. Montebugnoli, A. Lacava, N. Mohamadi, P. Brach del Prever, M. Polese, P. Bellavista, T. Melodia, "Managing the Managers: A Survey on AI-RAN Control, Orchestration, and Automation Towards 6G" Submitted to IEEE Communication Surveys and Tutorials, September 2026.

Honors and Awards

Best Presentation Award *December 2025*
IEEE ComSoc - CSIM 6G Industrial Dissemination Day [🌐]
◦ Paper: "AI-Enhanced RAN Infrastructure: Dynamic Resource Orchestration for Sustainable 6G Computing"

Best Paper Award *July 2023*
IEEE ISCC 2023 – 13th Workshop on Management of Cloud and Smart City Systems (MoCS) [🌐]
◦ Paper: "A Multi-Cloud Observability Support Based on ElasticSearch for Cloud-Native Smart Cities Services"

Tutorials

IEEE CCNC 2027 – Tutorial TUT-01 *January 9, 2027*
From Simulation to AI-Native Deployment: Building, Testing, and Operating Open RAN Control Applications [🌐]
◦ Half-day tutorial; sole in-person presenter, with L. Foschini and P. Bellavista

Research Projects

AutoRAN (Northeastern University) *2025*
NTIA Public Wireless Supply Chain Innovation Fund – Zero-touch integration and testing of multi-vendor cellular stacks
◦ Cloud-native and infrastructure-as-code components of the automation pipeline

RESTART SUPER Task *March 2023 - February 2026*
Next Generation EU Funds - PNRR National Project – Integration of optimization models in RAN and Core 5G [🌐]
◦ Integration of optimization models, virtualization, orchestration, and management solutions in 5G architecture
◦ Focus on RAN intelligent controllers and service orchestration

RESTART Industrial Networks (IN) Task *March 2023 - February 2026*
Next Generation EU Funds - PNRR National Project – Network architectures for Industrial IoT [🌐]
◦ Investigation of network architectures and computing infrastructures for IIoT applications
◦ Research on 5G integration with industrial applications

Thesis Supervision

Co-supervised 4 Master's theses and more than 20 Bachelor's theses and project activities in cloud continuum, observability, orchestration, and 5G networking.
Project Activities:

– "Polybench Isolation Technologies" – Alessio Benenati (Supervisor: L. Foschini)

Professional Service

Publication Chair *2026*
European Wireless (EW 2026), Rimini, Italy [🌐]

Web Co-Chair *2026*
31st IEEE Symposium on Computers and Communications (ISCC 2026) [🌐]

Web Co-Chair *2025*
IEEE CAMAD 2025 – Computer-Aided Modeling and Design of Communication Links and Networks [🌐]

Web Co-Chair *2025*
30th IEEE Symposium on Computers and Communications (ISCC 2025), Bologna, Italy [🌐]

Technical Program Committee:

– IEEE CCNC 2027 – TPC Member, Track T3: Testbeds, Experimentation and Datasets for Communications and Networking

– IWNC 2026 – TPC Member

– IEEE ISCC 2026 – TPC Reviewer

– MoCS 2026 Workshop at IEEE ISCC 2026 – TPC Member

– IEEE ICC 2026 Workshop WS-01 on Open RAN – TPC Member

– IEEE ISCC 2025 – TPC Reviewer

– MoCS 2025 Workshop at IEEE ISCC 2025 – TPC Member

– IEEE WCNC 2025 Workshop WS05 on Open Radio Access Networks – TPC Member

Peer Review Activity:
Conferences: IEEE ICC 2025, IEEE WCNC 2025, CNSM 2025, WiMob 2025, IEEE CCNC 2026
Journals: IEEE Transaction on Mobile Computing (2026), IEEE Transactions on Green Communications and Networking (TGCN) 2026, IEEE Transactions on Network and Service Management (TNSM) 2024-2026, IEEE Networking Letters 2026, IEEE Vehicular Technology Magazine 2026, Elsevier Computer Networks 2025-2026, Elsevier Computer Communications 2026, Elsevier Journal of Network and Computer Applications 2026, Elsevier Sustainable Computing: Informatics and Systems 2026, Elsevier MethodsX 2024

Skills & Abilities

Infrastructure Platforms: Kubernetes, Docker, Docker Compose, OpenStack, Terraform, Helm, Knative, OpenShift, Proxmox, Apache Kafka, RabbitMQ, MQTT, Ceph, ElasticSearch Stack (ELK), Prometheus, Grafana

5G & O-RAN Platforms: Near-RT RIC, Non-RT RIC, SMO, xApp/rApp Development, NS3, ns-O-RAN, OpenAirInterface (OAI), srsRAN, FlexRIC

Service Mesh & Orchestration: Istio, Istio Ambient Mesh, Linkerd, Cilium, TOSCA, Crossplane

DevOps & CI/CD: Git, GitHub, GitLab, Ansible, CI/CD Pipelines, ArgoCD, Jenkins

System Administration: Linux/Unix, Bash Scripting, VPN Configuration, Network Configuration, Virtualization (KVM, LXD)

Programming Languages: Python, C/C++, Java, Go, Rust, Bash, SQL

Web Technologies: HTML, CSS, JavaScript, React, Node.js, Spring, Spring Boot, Jakarta EE, RESTful APIs, WildFly

Database Systems: MySQL, MongoDB, ElasticSearch, PostgreSQL, InfluxDB

Machine Learning & Data: scikit-learn, Pandas, NumPy, TensorFlow, MLflow, MCP, RAG

HPC & Scientific Computing: ScaLAPACK, MPI, SLURM, Linear Algebra, Optimization Algorithms

Development Tools: VS Code, IntelliJ IDEA, Eclipse, vim, LaTeX

Other Tools: Adobe Creative Suite (InDesign, Illustrator, Photoshop), Wireshark, tcpdump

Conference Attendance

IEEE ICC 2026– Glasgow, United Kingdom *May 2026*
Cloud Native Days Italy 2026 (CNCF)– Bologna, Italy *May 2026*
Open6G + AI Workshop– Boston, MA, US *March 2026*
Next-Generation EU RESTART Project Plenary Meeting– Rome, Italy *January 2026*
30th IEEE ISCC– Bologna, Italy *July 2025*
IEEE ICC 2025– Montreal, Canada *June 2025*
Next-Generation EU RESTART Project Plenary Meeting– Palermo, Italy *January 2025*
IEEE GLOBECOM 2024– Cape Town, South Africa *December 2024*
IEEE HPSR 2024– Pisa, Italy *July 2024*
Lipari School on Advanced Networking Systems– Lipari, Italy *July 2024*
RESTART Plenary Workshop– Catania, Italy *July 2024*

IEEE MetroInd4.0&IoT 2024– Florence, Italy *May 2024*
5G&Beyond Conference– Rome, Italy *April 2024*
SC'23 – Supercomputing Conference– Denver, CO, USA *November 2023*
Next-Generation EU RESTART Tech Camp on 5G and O-RAN– Rome, Italy *September 2023*
SSIE Summer School– Bressanone, Italy *July 2023*
28th IEEE ISCC 2023– Tunis, Tunisia *July 2023*

Professional Memberships

IEEE– Graduate Student Member *2022 - Present*
IEEE Communications Society (ComSoc)– Student Member *2023 - Present*
ACM– Student Member *2023 - Present*
N²Women– Member *2025 - Present*

Additional Information

Languages: Italian (Native), English (Full Professional Proficiency – C1), Spanish (A1)
Soft Skills: Public Relations, Event Planning, Organizational Skills, Technical Writing, Scientific Presentations

In compliance with the GDPR and the Italian Legislative Decree no. 196 dated 30/06/2003, I hereby authorize the recipient to use and process my personal details contained in this document.